Certified local approximations to triangle-network correlations

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Abstract

We give an exactly symmetric finite triangle-local distribution with $P(A = B = C) = 41/144 > 1/4$, addressing the exact-symmetry question posed by Bäumer, Gitton, Kriváchy, Gisin, and Renner. We also provide a certification and interoperability layer for numerical local models. It exports independent source laws and response tables as portable rational witnesses, checks locality and squared residuals exactly, and adds a validated target enclosure to obtain a rigorous upper bound U on distance to the intended distribution. Selected development witnesses approximate noisy EJM and Renou correlations with U between 6×10^{-11} and 4×10^{-8}. A separate CPU experiment compares triangle-local, the public LHV-Net 2020 architecture, and localmodels on identical target arrays, with ten independent cold starts per method and target and 10- and 60-second budgets. Neural checkpoints receive fresh held-out evaluation with uncertainty. The software supports certification of numerical constructions without rerunning or trusting their search procedure.

1 An exactly symmetric triangle-local witness

The triangle network has three independent sources α, β, γ, shared by BC, CA, and AB, respectively. Local responses depend on the two sources incident to each party [1]. Bäumer et al. [6] ask whether a four-outcome triangle-local distribution with full party and simultaneous outcome-relabeling symmetry can have $P(A = B = C) > 1/4$. Their discussion includes approximately symmetric neural examples near 0.29. The following construction gives exact symmetry and an exact probability.

Put $O = \{0, 1, 2, 3\}$ and $S = \{(i, j) \in O^2 : i \neq j\}$. Choose α, β, γ independently and uniformly from the twelve elements of S. For $x = (x_1, x_2)$ and $y = (y_1, y_2)$ define

$$f(x, y) = \begin{cases} x_2, & x_2 \in \{y_1, y_2\}, \\ x_1, & \text{otherwise}, \end{cases} \qquad (A, B, C) = (f(\beta, \gamma), f(\gamma, \alpha), f(\alpha, \beta)).$$

Each response is deterministic and uses precisely its incident sources, so the resulting law is triangle-local. Enumerating the 12^3 equally likely source triples gives

$$P(a, b, c) = \frac{1}{1728} \begin{cases} 123, & |\{a, b, c\}| = 1, \\ 19, & |\{a, b, c\}| = 2, \\ 23, & |\{a, b, c\}| = 3. \end{cases} \tag{1}$$

There are 4, 36, and 24 output triples in these three classes, and $4(123) + 36(19) + 24(23) = 1728$. The formula depends only on the equality pattern; hence it is invariant under all party permutations and all simultaneous permutations of the four output labels. In particular,

$$P(A = B = C) = \frac{4(123)}{1728} = \frac{41}{144} = \frac{1}{4} + \frac{5}{144}.$$

This is an explicit answer to the existence question as stated in that paper, with full symmetry established exactly. The small integer enumeration and the complete finite response tables are included with the certificate. A separate revision-pinned Lean development records the continuous-source formulation; the computations reported here use the independent integer checker.

2 What a certificate establishes

A finite model consists of rational probability vectors $q_\alpha, q_\beta, q_\gamma$ and rational row-stochastic response tables $A^a_{jk}, B^b_{ki}, C^c_{ij}$. Its distribution is

$$Q(a,b,c) = \sum_{i,j,k} q_\alpha(i) q_\beta(j) q_\gamma(k) A^a_{jk} B^b_{ki} C^c_{ij}. \tag{2}$$

Checking nonnegative entries and unit row sums establishes valid response kernels and independent source laws. Clearing denominators in (2) reduces the full contraction to integer arithmetic. The checker recomputes all probabilities, normalization, symmetry, L_1 error, and squared Euclidean error; it does not trust an optimizer's success flag. Finite-cardinality methods are established in network locality [2, 4]. Here the deliverable is the exported, independently checkable object.

Write P for the intended target and $\widetilde{P}$ for the rational target stored with the witness. The checker computes $E = \|\widetilde{P} - Q\|_2^2 \in \mathbb{Q}_{\geq 0}$ exactly. A separate interval calculation supplies $\|P - \widetilde{P}\|_2 \leq \delta_{\text{target}}$. Thus

$$d_\triangle(P) := \inf_{R \in \mathcal{L}_\triangle} \|P - R\|_2 \leq \|P - Q\|_2 \leq \sqrt{E} + \delta_{\text{target}} \leq U. \tag{3}$$

Every displayed U is rounded upward. The δ_{target} values are outward-rounded interval upper bounds. Separate high-precision point evaluations are cross-checks, not the justification of those bounds. The interval formulas use 100 decimal digits; their binary endpoints are converted to exact rationals, followed by exact endpoint subtraction and an integer ceiling for the square root.

If $E = 0$, the stored target is local. Locality of the intended target follows when target identification also establishes $P = \widetilde{P}$. A positive U certifies a nearby local distribution, regardless of its size. A positive fit residual supplies neither a nonlocality proof nor a lower bound on $d_\triangle(P)$. The integer checker is tested software, not a formally verified implementation. A second standard-library audit uses rational variable elimination with a different contraction order and no package imports.

Target identity and noise placement

Decimal parameters denote their exact decimal rationals: for example, $u^2 = 0.85$ means $17/20$. Outputs use $a, b, c \in \{0, 1, 2, 3\}$ with index $16a + 4b + c$. For EJM, the entries for one, two, and three distinct labels are

$$\frac{1}{256}(4 + 9v + 9v^2 + 3v^3,\ 4 + v - 3v^2 - v^3,\ 4 - 3v + 3v^2 + v^3)$$

under independent source noise and

$$\frac{1}{256}(4 + 9v^2 + 12v^3,\ 4 + v^2 - 4v^3,\ 4 - 3v^2 + 4v^3)$$

under independent detector noise. Rational v gives exact rational targets, so their stored witnesses have $\delta_{\text{target}} = 0$. For Renou [7], the outcome basis is $|01\rangle, |10\rangle, u|00\rangle + \sqrt{1-u^2}|11\rangle, \sqrt{1-u^2}|00\rangle - u|11\rangle$, in cyclic qubit order, with three $|\phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ sources. For source visibility v, each independent edge instead carries

$$\rho_v = v|\phi^+\rangle\langle\phi^+| + (1-v)I_4/4, \qquad \rho_{\text{network}} = \rho_v^{\otimes 3}.$$

This is the source-noise convention in Table 1, including $v = 0.89$. Independent detector noise applies $K_v^{\otimes 3}$, where $K_v = vI + (1-v)J/4$; it is different from global white-noise mixing. General Renou targets require algebraic enclosures.

Transport to lower detector visibility

Let $0 \leq v_L \leq v_H$ with $v_H > 0$, and $r = v_L/v_H$. Applying K_r to each response row preserves locality and gives $Q_L = K_r^{\otimes 3} Q_H$. Since $K_r K_{v_H} = K_{v_L}$, and the eigenvalues of the symmetric channel K_r are $1, r, r, r$, its Euclidean operator norm is one. Therefore

$$\|Q_L - P(v_L)\|_2 \leq \|Q_H - P(v_H)\|_2 \leq U_H.$$

An exact high-visibility target model yields exact lower-visibility models; an approximate one yields approximations with this controlled error.

3 Selected development witnesses

Table 1 reports selected witnesses from the complete development campaign, including fitted seeds, continuation and polishing. They are separate from the cold-start comparison below; their earlier preparation is not free initialization for that experiment. Each bound follows from (3).

Target	$\sqrt{E}$	δ_{target}	U
Exactly symmetric construction	0	0	0
Renou noiseless, $u^2 = 0.50$	0	0	0
EJM source, $v = 0.80$	5.77×10^{-11}	0	5.774×10^{-11}
EJM detector, $v = 0.86$	1.51×10^{-9}	0	1.51×10^{-9}
Renou source, $v = 0.89$	3.27×10^{-8}	5.227×10^{-13}	3.27×10^{-8}
Renou detector, $v = 0.93$	8.21×10^{-9}	2.238×10^{-14}	8.208×10^{-9}
EJM noiseless	0.0483	0	0.04835
Renou noiseless, $u^2 = 0.85$	0.00224	2.782×10^{-14}	0.002241

Table 1: Selected development witnesses. Residuals $\sqrt{E}$ are rounded approximations; both enclosure columns are rounded upward. Renou noise rows have $u^2 = 0.85$. An exact zero is stated only where target identity is exact.

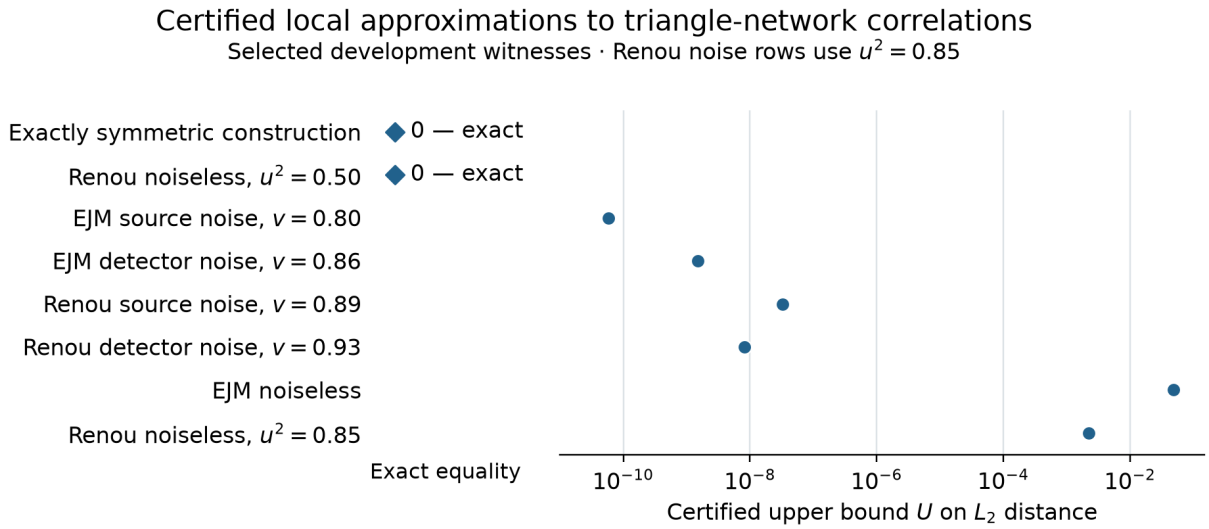


Figure 1: Certified bounds for the selected distributions. Exact zeros occupy a separate labeled column; positive bounds use a logarithmic axis. Each point represents an exactly local model. Points are not interpolated into a locality boundary.

At $v = 1$, source and detector noise give the same noiseless target. It is counted once per quantum family. Different continuation paths can find different candidate residuals for that one target; these are search outcomes, not two physical distances. The noiseless Renou controls $u^2 = 1/2$ and $u^2 = 1$ change the measurement parameter and are distinct from visibility $v = 1$ at $u^2 = 0.85$.

A compact Renou witness: $u^2 = 0.85$, detector $v = 0.93$. The original-target bound is $U \leq 8.208 \times 10^{-9}$. The compressed bundle occupies 6,007 bytes; median exact checking time is 3.63 ms over eleven checks. Exact zero-weight removal changes source sizes from $(12, 12, 12)$ to $(8, 9, 2)$ without changing the output distribution. This is exact trimming; minimum cardinality is not asserted. The checking timer includes model parsing, simplex checks, exact contraction, distances and hashes, with the bundle already decompressed in memory.

4 A matched cold-start comparison

The experiment compares triangle-local, LHV-Net-2020 [3], and localmodels [4] on three declared targets: EJM source $v = 0.80$, EJM detector $v = 0.86$, and Renou detector $u^2 = 0.85$, $v = 0.93$. Each method receives the same stored 64-entry double-precision array T, with its ordering, exact physical parameters and SHA-256 recorded. All reported scores are recomputed in double precision against that array, including neural outputs. The training loss may differ. In particular, localmodels reports $\sqrt{\mathrm{SSE}/64}$ internally: we explicitly verify

$$L_2 = \sqrt{\mathrm{SSE}} = 8\,\mathrm{RMSE} \tag{4}$$

from every saved output, rather than reading its internal error field.

There are ten independent runs per method–target pair, each observed at 10 and 60 seconds. No method receives an analytic control, recognized target, fitted seed, or neighboring checkpoint. The run order is deterministically shuffled. One solver process runs at a time on an Apple M2 Pro CPU with 16 GiB memory and one numerical-library thread; GPUs are disabled. The timing begins with initialization after loading the target and common libraries. It includes model construction, optimization, and checkpoint selection. Held-out evaluation, export, and certification are additional costs.

Both finite methods use seven symbols per source, with unrestricted stochastic responses and free independent source weights. Triangle-local alternates L-BFGS (400-iteration cap), convex polishing (120 sweeps, 120 inner steps), and two perturbation searches, restarting if the cycle completes. It calls the hybrid routine directly, bypassing target recognition. Localmodels uses its public constrained trust-region optimizer, one random trial per call and zero outer error tolerance, restarting on return. The benchmark retains the best feasible output evaluated before each deadline; infeasible optimizer trial points are excluded.

LHV-Net uses the pinned public response routing and architecture with three hidden tanh layers of width 16, batch size 256, and Adadelta learning rate 2 on the KL loss. An eager TensorFlow wrapper replaces obsolete Keras generator calls. This is a small CPU configuration of the public code, not a replication of every architecture and training budget used in the 2020 paper. Neural checkpoints minimize a fixed validation-set residual, with all validation charged to search time. Final testing is separate, as detailed below.

Target	Method	10 s L_2	60 s L_2	Successes	
				10 s	60 s
EJM source	triangle-local	$0.0109[0.0063, 0.0125]$	$0.00544[0.00431, 0.00658]$	0/10	0/10
	LHV-Net-2020	$0.0722[0.0669, 0.0767]$	$0.0511[0.0506, 0.053]$	0/10	0/10
	localmodels	$0.11[0.103, 0.118]$	$0.0518[0.0494, 0.0538]$	0/10	0/10
EJM detector	triangle-local	$0.017[0.014, 0.0233]$	$0.00674[0.00508, 0.0075]$	0/10	0/10
	LHV-Net-2020	$0.0703[0.0668, 0.0745]$	$0.0499[0.0488, 0.0505]$	0/10	0/10
	localmodels	$0.103[0.102, 0.105]$	$0.0496[0.0475, 0.0512]$	0/10	0/10
Renou detector	triangle-local	$(2.86[1.22, 7.64]) \times 10^{-5}$	$(3.91[1.67, 5.59]) \times 10^{-6}$	10/10	10/10
	LHV-Net-2020	$0.0279[0.0268, 0.0295]$	$0.0172[0.0154, 0.0188]$	0/10	0/10
	localmodels	$0.122[0.0258, 0.216]$	$0.00792[0.00715, 0.00919]$	0/10	0/10

Table 2: Cold-start L_2 to the common target: median [first, third quartile] over ten runs. LHV-Net-2020 uses the small CPU configuration with an eager TensorFlow wrapper. Success counts use the predeclared threshold 10^{-3}. For neural outputs a success requires the upper endpoint of the held-out 95% interval to meet the threshold. The two budgets are checkpoints of the same ten runs, not twenty independent trials.

At 60 seconds the declared threshold was met by 10/30 triangle-local runs, 0/30 neural runs, and 0/30 localmodels runs. These counts summarize the specified targets and settings; ten trials per cell do not establish population-level reliability or a universal ranking.

5 Evaluation precision and interoperability

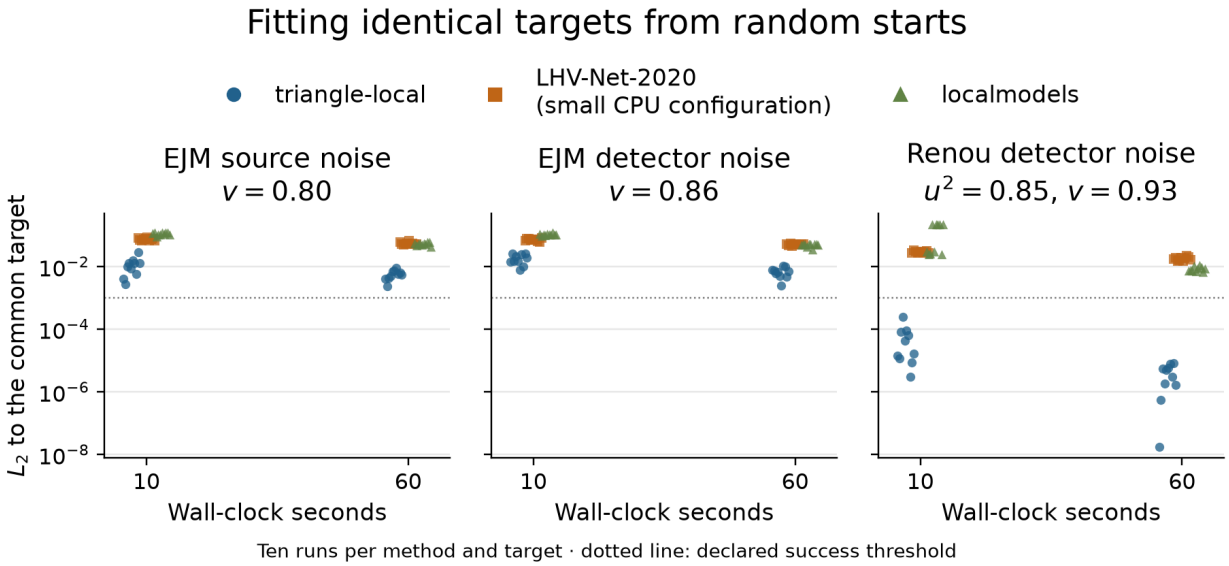


Figure 2: All ten individual runs at each completed budget, with horizontal jitter for legibility. LHV-Net-2020 uses the small CPU configuration with an eager TensorFlow wrapper. Neural error bars are held-out approximate 95% intervals; finite models are evaluated by full contraction. Neural intervals are usually narrower than the markers. Only the two measured budgets are shown. The vertical axis is an operational L_2 score, not a certificate bound.

For a fixed neural checkpoint let X be its conditional-product probability vector, $Q = \mathbb{E}X$, and $\widehat{Q}_N = N^{-1}\sum_{i=1}^{N} X_i$. Fresh independent source triples give

$$\mathbb{E}\|\widehat{Q}_N - T\|_2^2 = \|Q - T\|_2^2 + \frac{\operatorname{tr}\operatorname{Cov}(X)}{N}.$$

Each selected checkpoint receives $N = 2^{20}$ fresh samples in 32 blocks, after both checkpoint choices are frozen. These samples are independent of training and validation and never select a checkpoint. Tables and Figure 2 report the raw score $s = \|\widehat{Q}_N - T\|_2$, without variance correction or truncation. For the $B = 32$ block means $\overline{X}_b$, we save

$$\widehat{V} = \frac{1}{B(B-1)}\sum_{b=1}^{B}\|\overline{X}_b - \widehat{Q}_N\|_2^2.$$

The separate diagnostic $\sqrt{\max(0, s^2 - \widehat{V})}$ and estimated RMS sampling error $\sqrt{\widehat{V}}$ are not the plotted score. Each of 2,000 bootstrap resamples draws B block means with replacement and recomputes the raw norm s^*. If q_p is the empirical p-quantile of these norms (NumPy's default linear interpolation), the displayed basic 95% interval is

$$[\max(0, 2s - q_{0.975}),\ \max(0, 2s - q_{0.025})].$$

Thus both interval endpoints are clipped at zero. These are approximate statistical intervals, not rigorous enclosures. Finite-model outputs use full contractions. Held-out evaluation took a median 0.87 seconds per checkpoint. Estimated Monte Carlo L_2 RMS errors ranged from 0.000419 to 0.000733; the median interval half-width was 0.000226. There were 0 threshold-straddling neural intervals.

From a saved neural checkpoint to a rational witness

We reload the preselected 60-second Renou checkpoint from repetition zero. Four uniform support points are chosen independently for each source; every incident pair is evaluated and the full Cartesian product is contracted. Rationalization gives a finite triangle-local witness. An empirical average over sampled triples would not enforce a product source law, and the extracted finite model need not equal the continuous network.

Object	Accuracy	Additional cost
Continuous neural checkpoint	$L_2 = 0.0165$	60.0 s training/selection
Extracted finite model	$U \leq 0.03136$	0.29 s extraction
Extracted model + polish	$U \leq 0.0006423$	2.00 s polish
Random controls + polish	0.001602 median U	2 s each; ten controls

Table 3: Native neural evaluation, finite extraction, and subsequent polishing are distinct objects. Finite rows report the original-target bound U. Random controls use the same four-symbol cardinalities and two-second polishing budget. Training, extraction, rationalization and checking costs are recorded separately.

6 Scope, context, and reproducibility

The practical contribution is a certification and interoperability layer: portable rational source distributions and response tables, rigorous accounting for target approximation, compact representations, and a small checker. LHV-Net already defines generative local models, and localmodels already provides finite response tables. Both can supply numerical constructions for this certificate pipeline. The finite-model benchmark exports from both finite solvers are rationalized and independently checked after timing; the native search comparison remains separate from that certificate-production cost.

The polished extraction lies below the median of the ten random controls, whose bounds range from 0.0005051 to 0.008027. This is one preselected neural checkpoint, so it establishes a reproducible conversion example rather than a general initialization advantage. Rationalization and verification took 0.005 seconds before polishing and 0.004 seconds afterwards.

The numerical search uses full finite contractions and nonconvex optimization. Fixing five factors makes the sixth factor's squared-loss subproblem convex, but finite inner iteration caps remain part of the method. Recognizers, lifted analytic seeds, and continuation are useful optional workflows whose preparation costs must be accounted for. The earlier configuration study is descriptive: its runs have different total budgets and do not isolate causal effects.

Configuration	$\sqrt{E}$	Added time (s)	Total (s)
Blind L-BFGS, 12 symbols	0.0198	–	4.57
Seeded L-BFGS	0.00543	–	3.76
Seeded L-BFGS + fixed-weight polish	0.00163	5.80	9.56
Seeded L-BFGS + free-weight polish	0.000407	8.23	11.99
Seeded hybrid + perturbations	5.06×10^{-10}	–	27.98

Table 4: Illustrative earlier EJM source configurations at $v = 0.80$. Both polishing variants branch from the same seeded L-BFGS model. Additional and total times are separated. The residual is to the stored target.

Kriváchy et al.'s fitted noise curves are historical context, not measured baseline outcomes or certified lower bounds. The present CPU experiment does not compare GPUs, long continuation workflows, or the adaptive PyTorch toolbox of Girardin et al. [5]; it establishes no unrestricted state-of-the-art optimizer ranking. Gitton and Renner's exact-arithmetic nonclassicality proof for ideal EJM [8] is complementary to our constructive upper bounds. An exclusion alone is not a numerical distance lower bound. The unrestricted optimum is not determined here.

Artifacts and checking

The triangle-local repository contains the public source and prior certificate collection. The accompanying revision-labelled archive contains this note, the checker, tests, exact witnesses, target arrays, frozen protocol, all raw outcomes, saved neural weights, and a content-hash manifest. It is a local reproducibility artifact; no unpushed revision is presented as a published release. External implementations are identified by immutable source revisions and require separate installation. The archived target-generator spelling patch is applied only in a downstream reference check.

The exact-symmetry enumeration is reproducible without running an optimizer. The test suite covers analytical derivatives, simplex constraints, changed certificates, detector-channel composition, zero-weight trimming, and outward rounding. The independent rational audit checks every selected witness and every newly exported finite checkpoint. Raw distributions and target hashes also allow the three-way scores to be recomputed independently of optimizer logs. The final suite passed 68 tests. The Round 4 audit recomputed all 180 native outcomes and checked all 120 finite checkpoint certificates; the earlier selected witness collection was also rechecked. The serial comparison took 1.56 hours including process startup and held-out evaluation.

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